

**BOM -- EVB-USB5744\_A1****RoHS****EVB-USB5744, A0 Silicon, 4-port Evalutaion Board for QFN56 package.**

Assembly Revision A1

(Truffle)

Revised: 20150924

| Item | Qty | Qty<br>Pop'd | Reference Designator(s)                                                                                                                                                | Description                                                                                   | Manufacturer           | Manufacturer Part Number | Notes                                           |
|------|-----|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------|--------------------------|-------------------------------------------------|
|      |     |              | C1,C2,C3,C4,C5,C6,C7,C8,C9,C10,C11,C12,C18,<br>C19,C20,C23,C27,C30,C31,C34,C37,C38,C40,C41,<br>C42,C43,C44,C47,C50,C51,C54,C55,C56,C57,C58,<br>C59,C60,C61,C66,C67,C68 | Capacitor, 0.1uF, 10%, 25V, X5R, 0402                                                         | Murata                 | GRM155R61E104KA87D       |                                                 |
| 1    | 41  | 41           | C13,C14,C15,C16,C17                                                                                                                                                    | Capacitor, Low ESR, 150uF, 6.3VDC, 20%, Aluminum, Radial-SMT, 5mm x 5.7mm                     | Lelon                  | VZS151M0J1TR-0506        |                                                 |
| 2    | 5   | 5            | C24,C25,C26,C48,C49,C52,C62,C64                                                                                                                                        | Capacitor, 1000pF, 50V, 10%, X7R, 0402                                                        | Murata                 | GRM155R71H102KA01D       |                                                 |
| 3    | 8   | 8            | C28,C39                                                                                                                                                                | Capacitor, 10uF, 6.3VDC, 20%, X5R, 0603                                                       | Murata                 | GRM188R60J106ME47D       |                                                 |
| 4    | 2   | 2            | C29,C33                                                                                                                                                                | Capacitor, 16pF, 50V, 2%, NPO, 0402                                                           | Murata                 | GRM1555C1H160GA01D       |                                                 |
| 5    | 2   | 2            | C35                                                                                                                                                                    | Capacitor, 1.0uF, 16VDC, 10%, X5R, 0603                                                       | Murata                 | GRM188R61C105KA93D       |                                                 |
| 6    | 1   | 1            | C36,C45                                                                                                                                                                | Capacitor, 4.7uF, 6.3VDC, 20%, X5R, 0603                                                      | Murata                 | GRM188R60J475KE19D       |                                                 |
| 7    | 2   | 2            | C46,C63                                                                                                                                                                | Capacitor, 10uF, 25 VDC, 10%, 0805                                                            | Murata                 | GRM21BR61E106KA73L       |                                                 |
| 8    | 2   | 2            | C53                                                                                                                                                                    | Capacitor, 2.2uF, 6.3VDC, 10%, X5R, 0603                                                      | Murata                 | GRM185R60J225KE26D       |                                                 |
| 9    | 1   | 1            | C65                                                                                                                                                                    | Capacitor, 0.01uF, 25V, 10%, X7R, 0402                                                        | Panasonic              | ECJ-0EB1C103K            |                                                 |
| 10   | 1   | 1            | D1,D2,D3,D4                                                                                                                                                            | LED, Green, 3mm, Diffused, 0.2" CL-vert, TH, Right Angle                                      | Lumex                  | SSF-LXH103GD             |                                                 |
| 11   | 4   | 4            | D5,D7,D8,D9,D10,D13                                                                                                                                                    | LED, Bright Green, 0603                                                                       | Liteon                 | LTST-C191KGKT            |                                                 |
| 12   | 6   | 6            | D11,D14                                                                                                                                                                | LED, Blue, 0603                                                                               | Stanley Electric       | DB1111C-TR               |                                                 |
| 13   | 2   | 2            | D12                                                                                                                                                                    | LED, Red, 0603                                                                                | Stanley Electric       | BR1111C-TR               |                                                 |
| 14   | 1   | 1            | D30,D31                                                                                                                                                                | Diode, MMBD914LT, Fast Switching, 100VDC, 200mA, SOT-23                                       | On Semiconductors      | MMBD914LT1               |                                                 |
| 15   | 2   | 2            | FB3,FB4                                                                                                                                                                | Ferrite Bead, 220 Ohm, 2A, 0.05DCR, 0603                                                      | Murata                 | BLM18EG221SN1D           |                                                 |
| 16   | 2   | 2            | R1,R2,R3,R4,R5,R6,R7,R8,R66                                                                                                                                            | Resistor, 330, 5%, 1/16W, 0603                                                                | Panasonic              | ERJ-3GEYJ331V            |                                                 |
| 17   | 9   | 9            | R9,R10,R14,R24,R25,R26,R27,R55                                                                                                                                         | Resistor, ZERO, 0.1W, 0603                                                                    | Panasonic              | ERJ-3GEY0R00V            |                                                 |
| 18   | 8   | 8            | R11                                                                                                                                                                    | Resistor, 12.0K, 1%, 1/16W, 0603                                                              | Panasonic              | ERJ-3EKF1202V            |                                                 |
| 19   | 1   | 1            | R12,R28,R29,R30,R31,R32,R57,R63,R67                                                                                                                                    | Resistor, 1K, 5%, 1/16W, 0603                                                                 | Panasonic              | ERJ-3GEYJ102V            |                                                 |
| 20   | 9   | 9            | R13                                                                                                                                                                    | Resistor, 47K, 5%, 1/16W, 0603                                                                | Yageo America          | 9C06031A4702JLHFT        |                                                 |
| 21   | 1   | 1            | R15                                                                                                                                                                    | Resistor, 21.0K, 1%, 1/10W, 0603                                                              | Panasonic              | ERJ-3EKF2102V            |                                                 |
| 22   | 1   | 1            | R16                                                                                                                                                                    | Resistor, 150K, 1%, 1/16W, 0603                                                               | Panasonic              | ERJ-3EKF1503V            |                                                 |
| 23   | 1   | 1            | R20,R21,R22,R23                                                                                                                                                        | Resistor, 100, 1%, 1/10W, 0603                                                                | Panasonic              | ERJ-3EKF1000V            |                                                 |
| 24   | 4   | 4            | R33                                                                                                                                                                    | Resistor, 1.91K, 1%, 1/10W, 0603                                                              | Stackpole Electronics  | RMCF0603FT1K91           |                                                 |
| 25   | 1   | 1            | R35,R36,R41,R43                                                                                                                                                        | Resistor, 200K, 5%, 1/10W, 0603                                                               | Vishay/Dale            | CRCW0603200KJNEB         |                                                 |
| 26   | 4   | 4            | R37,R38,R42,R44,R45,R58,R61,R64                                                                                                                                        | Resistor, 10K, 5%, 1/16W, 0603                                                                | Panasonic              | ERJ-3GEYJ103V            |                                                 |
| 27   | 8   | 8            | R39,R40,R46,R47                                                                                                                                                        | Resistor, 10, 5%, 1/16W, 0603                                                                 | Yageo America          | 9C06031A10R0JLHFT        |                                                 |
| 28   | 4   | 4            | R53,R54,R60,R62,R65                                                                                                                                                    | Resistor, 100K, 5%, 1/16W, 0603                                                               | Panasonic              | ERJ-3GEYJ104V            |                                                 |
| 29   | 5   | 5            | R56                                                                                                                                                                    | Resistor, 255, 1%, 1/10W, 0603                                                                | Panasonic              | ERJ-3EKF2550V            |                                                 |
| 30   | 1   | 1            | R59                                                                                                                                                                    | Resistor, 20, 5%, 1/10W, 0603                                                                 | Panasonic              | ERJ-3GEYJ200V            |                                                 |
| 31   | 1   | 1            | J10,J11                                                                                                                                                                | Header, 1 x 2, 0.1 Inch, Vertical                                                             |                        |                          |                                                 |
| 32   | 2   | 2            | J12                                                                                                                                                                    | Receptacle, USB 3.0, Style MicroB, Right Angle, SMT, with TID#, TH Tabs                       | Hirose Electric Co Ltd | ZX360D-B-10P             |                                                 |
| 33   | 1   | 1            | J1,J2,J3,J4                                                                                                                                                            | Receptacle, USB 3.0, Style A, Right Angle, Through-hole                                       | Amphenol               | GSB311131HR              |                                                 |
| 34   | 4   | 4            | J9                                                                                                                                                                     | Connector, Power Jack, 2.5 mm x 5.5 mm, 12 V, 4 A, Right Angle, TH                            | Cui Stack              | PJ-002BH                 |                                                 |
| 35   | 1   | 1            | SW1,SW2,SW3,SW4,SW7                                                                                                                                                    | Switch, DPDT, Slide, Sub-Mini, Top Actuator, TH                                               | C&K                    | JS202011CQN              |                                                 |
| 36   | 5   | 5            | SW5,SW6                                                                                                                                                                | Switch Array, SPST, 6 Position, SIP-7                                                         | CK                     | SPA06B                   |                                                 |
| 37   | 2   | 2            | SW8                                                                                                                                                                    | Switch, Momentary, SPST, 50mA, J-lead, NO, MicroMini                                          | C&K Components         | PTS810 SJM 250 SMTR LFS  |                                                 |
| 38   | 1   | 1            | U0                                                                                                                                                                     | IC, USB5744 USB 3.0 Hub, 4 port, QFN56                                                        | Microchip              | USB5744_A0               |                                                 |
| 39   | 1   | 1            | U1                                                                                                                                                                     | IC, DC-DC Converter Module, 0.591-6 Vout, ~12 Vin, 0.591-6 VDC out, 3A, 5 pin SIP, 0.41" Wide | Murata                 | OKR-T/3-W12-C            |                                                 |
| 40   | 1   | 1            | U3                                                                                                                                                                     | IC, DC-DC Converter Module, 0.591-6 Vout, ~12 Vin, 0.591-6 VDC out, 6A, 5 pin SIP, 0.41" Wide | Murata                 | OKR-T/6-W12-C            |                                                 |
| 41   | 1   | 1            | U2                                                                                                                                                                     | IC, MCP1725-ADJE/SN, LDO Regulator, Adj., 500 mA, SOIC8                                       | Microchip              | MCP1725(T)-ADJE/SN       |                                                 |
| 42   | 1   | 1            | U4,U5,U6,U7                                                                                                                                                            | IC, MIC2009YML, Power Distribution Switch, MLF-6                                              | Microchip              | MIC2009YML               |                                                 |
| 43   | 4   | 4            | U8                                                                                                                                                                     | IC, Flash, SPI, SST26VF016B, 16Mb (2M x 8), 2.7V-3.6V, 104MHz (Quad Read), SO8                | Microchip              | SST26VF016BT-104I/SN     |                                                 |
| 44   | 1   | 1            | U9,U10,U12                                                                                                                                                             | IC, 74LVC1G14, Inverter, Shottky, DCK                                                         | TI                     | SN74LVC1G14DCKR          |                                                 |
| 45   | 3   | 3            | U11                                                                                                                                                                    | IC, PI3B3257, Quad 2:1 Mux, 3.3V, QSOP-16                                                     | Pericom                | PI3B3257QE               |                                                 |
| 46   | 1   | 1            | Y1                                                                                                                                                                     | Crystal, 25.000MHz, 30ppm, 10pF, SMT 3.2MM X 2.5MM                                            | Abraco                 | ABM8G-25.000MHZ-B4Y-T    |                                                 |
| 47   | 1   | 1            | SHUNT1                                                                                                                                                                 | Shunt, Insulated, 0.1 Inch                                                                    | TE Connectivity        | 881545-2                 | Install across J10 pins 1 and 2.                |
| 48   | 2   | 2            | -none                                                                                                                                                                  | Foot, Silicone Rubber, Adhesive, Clear, Cylindrical, .500"x.250"                              | Bumper Specialties     | RBS-6                    | Place at each Board corner on the bottom side   |
| 49   | 4   | 4            | LBL-SERNO                                                                                                                                                              | Label, Serial Number, Laminated, 250 x 800                                                    | Brother                | 1/4" x 0.8"              | Place in white S/N box on Bottom Side of board. |
| 50   | 1   | 1            | LBL-ASSY1                                                                                                                                                              | Label, Assembly Number, Laminated, 250 x 800, "EVB-USB5744_A1"                                | Brother                | 1/4" x 0.8"              | Place in white Assy box on Top Side of board    |
| 51   | 1   | 1            | PCB Fab                                                                                                                                                                | PCB, Truffle (EVB-USB5744), Rev. A                                                            | Truffle A              |                          |                                                 |

|    |   |   |         |                                                        |           |                    |                                 |
|----|---|---|---------|--------------------------------------------------------|-----------|--------------------|---------------------------------|
| 53 | 2 | 0 | C21,C22 | Capacitor, 4.7uF, 6.3VDC, 20%, X5R, 0603               | Murata    | GRM188R60J475KE19D | Do Not Populate                 |
| 54 | 1 | 0 | C32     | Capacitor, 0.1uF, 10%, 25V, X5R, 0402                  | Murata    | GRM155R61E104KA87D | Do Not Populate                 |
| 55 | 1 | 0 | D6      | LED, Red, 3mm, Diffused, 0.2" CL-vert, TH, Right Angle | Lumex     | SSF-LXH103R        | Do Not Populate                 |
| 56 | 2 | 0 | FB1,FB2 | Ferrite Bead, 220 Ohm, 2A, 0.05DCR, 0603               | Murata    | BLM18EG221SN1D     | Do Not Populate                 |
| 57 | 1 | 0 | R52     | Resistor, ZERO, 0.1W, 0603                             | Panasonic | ERJ-3GEY0R00V      | Do Not Populate                 |
| 58 | 1 | 0 | R34     | Resistor, 29.4K, 1%, 1/10W, 0603                       | Panasonic | ERJ-3EKF2942V      | Do Not Populate                 |
| 59 | 1 | 0 | J6      | Header, 1 x 2, 0.1 Inch, Vertical                      | FCI       | 68000-236HLF       | Do Not Populate                 |
| 60 | 1 | 0 | J8      | Header, 2 x 5, 0.1 Inch, Vertical                      | AMP       | 146256-5           | Do Not Populate and Mask Off J8 |